

A Simpler Model Confirms the Wildfire Effect

The result

A **simpler** approach — one model trained on *all* the data (wildfire + clean), compared against a clean-only model — gives **the same answer** as the two-model method: wildfire raises CCN by about **16%** during wildfire periods (Figure 1). It recovers this with **no special handling**, and giving the wildfire data extra weight **did not change the answer** (Figure 2). The all-data model is also somewhat better calibrated than the wildfire-only model, since it trains on more data.

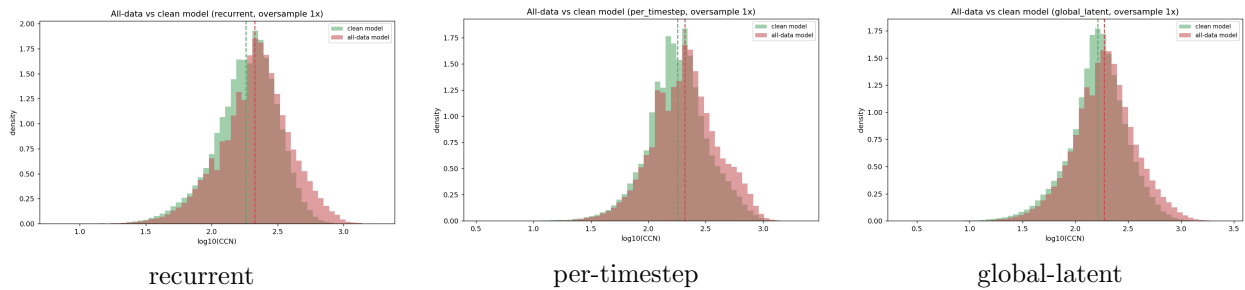


Figure 1: The all-data model (red) vs. the clean model (green) at wildfire-period weather, for all three model types (no oversampling). In every case the all-data model already sits at higher CCN — the wildfire effect is present, not washed out.

How the plots were made

Exactly as in the two-model report, both models are run *at the same weather*: the weather from the **held-out wildfire periods**. The **red** curve is the *all-data* model’s predicted CCN for that weather; the **green** curve is the *clean* model’s prediction for the same weather (what CCN would have been without wildfire). Both use the same wildfire-period weather as input; only the model differs.

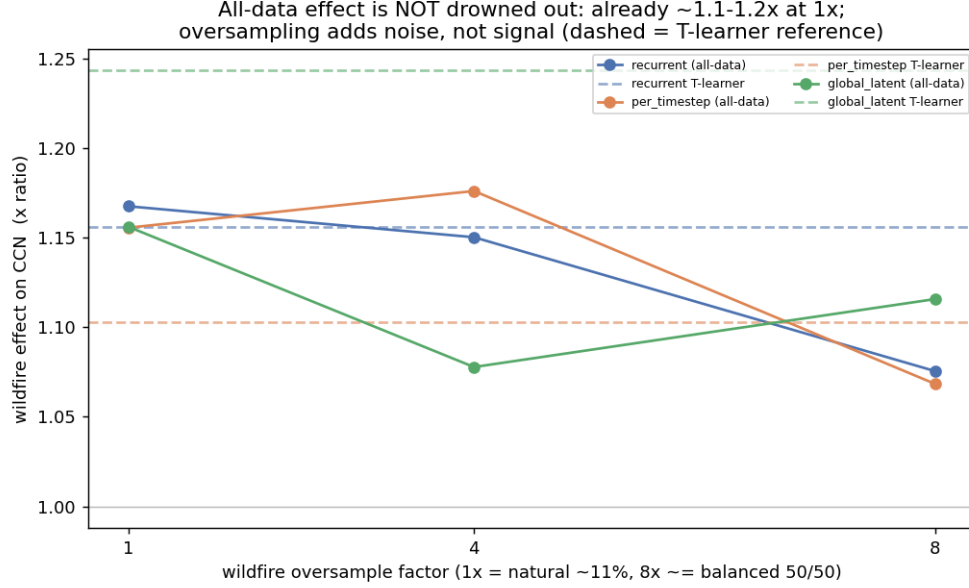


Figure 2: Effect from the all-data model (solid) vs. the two-model method (dashed), for three model types. They agree, and giving the wildfire data more weight (left to right) does not move the answer.

The three model types

All three make the LSTM output a *range* of possible CCN values (rather than a single guess) by injecting Gaussian noise; they differ in where the noise enters and how it varies across the trajectory:

- **Per-timestep:** a standard LSTM with a random noise vector concatenated onto the weather inputs, redrawn *fresh and independent at every timestep*.
- **Global-latent:** the *same* setup, except the noise vector is drawn *once per trajectory* and held fixed across all timesteps (one shared “latent code”).
- **Recurrent:** unlike the other two, the noise is *added* to the LSTM’s *hidden state* (not concatenated to the inputs) at each step, so the randomness *accumulates* through the sequence.

Why the simpler model might work here (a hypothesis)

We are **not sure** why the all-data model isn’t swamped by the clean majority ($\sim 89\%$ of the data). One *plausible* explanation: wildfire and clean periods tend to occur in *different weather*, so the model may be able to tell them apart and lean on the wildfire examples at wildfire-weather rather than averaging everything together — which would also explain why adding weight to the wildfire data changed little. We have not verified this mechanism; it is a hypothesis consistent with the results, not a demonstrated cause.

What to keep in mind

The small differences across settings are within noise (no error bars yet). If the weather-separation explanation above is right, the simpler model would be a good default only where that separation holds; if wildfire and clean conditions overlapped more, it could start blending them.

Bottom line

Both methods agree: wildfire adds roughly **15–20% more CCN**, and the simpler all-data model is enough for this dataset.